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Limit Points

1.1 Limit Points and Closure

Definition 1.1.1

Let A be a subset of a topological space $(X, \mathcal{T})$. A point $x \in X$ is said to be a **limit point** (or **accumulation point** or **cluster point**) of A if every open set, U , containing x contains a point of A different from x .

Note.

A 的极限点 x 是不能与 $A \setminus \{x\}$ “分离开”的点. 也就是说在每一个开集里面, x 和 $A \setminus \{x\}$ 要么都有在里面的点, 要么都没有.

Proposition 1.1.2

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is closed in $(X, \mathcal{T})$ if and only if A contains all of its limit points.

Proof sketch.

如果集合 A 是闭的, 那么 A 外面的点 x 都被落在 A 外面的一个开集 U 给包住了, 那么就导致了 x 不是极限点.

如果 A 包含了所有极限点, 那么 A 外面的点 x 都不是极限点, 从而 x 一定能被与 A 无交的一个开集给包住.

Proof. If A contains all of its limit points. Take any $x \in A^c$, then x is not a limit point of A . There exists open set U , containing x contains no point of A . Then $U \subseteq A^c$, which implies that A^c is an open set. Thus A is a closed set.

Now, let A be a closed set. Let x be any limit point of A . If $x \notin A$, then there is a open set $U \subseteq A^c$ such that $x \in U$. By the definition of the limit point, there is a point y of A different from x belongs to U , which is a contradiction to $U \cap A = \emptyset$. $\square$

Proposition 1.1.3

Let A be a subset of a topological space $(X, \mathcal{T})$ and A' the set of all limit points of A . Then $A \cup A'$ is a closed set.

Remark.

The proof shows that : If U is an open set, and A an arbitrary set. Then $U \cap A = \emptyset$ implies that $U \cap A' = \emptyset$.

Proof. Let $p \in X \setminus (A \cup A')$, then p neither a point in A nor a limit point of A . Thus there exists a open set U , such that $p \in U$, $U \cap A = \emptyset$. We now claim that $U \cap A' = \emptyset$, otherwise, we take $q \in U \cap A'$. Then U contains a point of A different from q , which is a contradiction to $U \cap A = \emptyset$. Thus we have

$$U \cap (A \cup A') = \emptyset \implies p \in U \subseteq X \setminus (A \cup A')$$

Then $X \setminus (A \cup A')$ is an open set. □

Definition 1.1.4

Let A be a subset of a topological space $(X, \mathcal{T})$. Then the set $A \cup A'$ consisting of A and all its limit points is called the **closure of A** and is denoted by $\bar{A}$.

Definition 1.1.5

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is said to be **dense** in X or **everywhere dense** in X if $\bar{A} = X$.

Proposition 1.1.6

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is dense in X if and only if every non-empty open subset of X intersects A non-trivially (that is, if $U \in \mathcal{T}$ and $U \neq \emptyset$ then $A \cap U \neq \emptyset$.)

Proof. If A is dense in X , then $\bar{A} = X$. Take any non-empty open subset U of X . If $A \cap U = \emptyset$, then $A' \cap U = \emptyset$ from Proposition 1.1.3 Remark . Thus $U \cap \bar{A} = \emptyset$, then $U \cap X = \emptyset$, which is a contradiction. Thus $A \cap U \neq \emptyset$.

Conversely, if for any $U \in \mathcal{T}$, and $U \neq \emptyset$, we have $A \cap U \neq \emptyset$. Take any $x \in X$. If $x \notin A$, then for any open set $V \in \mathcal{T}$, containing x , we have

$A \cap V \neq \emptyset$. And we have $V \cap (A \setminus \{x\}) \neq \emptyset$. Thus x is a limit point of A . It follows that $x \in A \cup A' = \bar{A}$, $X \subseteq \bar{A}$. Then $X = \bar{A}$. $\square$